

03 Host a Web Site with TLS

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Goals

- Learn some Linux command line skills
- Learn how to add a certificate to a web server
- Understand TLS in practice

Steps

1. Get a VM
2. Get a web server
3. Run web server
4. Get certificate and private key
5. Install certificate and key in web server
6. Verify TLS functionality

Submission

Your submission should include the following items:

1. Screen shot of your browser having a successful HTTPS connection to your web server
2. Your web server certificate
3. Your web server certificate chain

In addition, please include your answers to the following questions.

1. What web server and version did you run?
2. What TLS implementation and version does your web server use? How did you find this information?
3. (Extra credit) What TLS version and cipher suite is used when you connect from your browser? How did you find out?
4. What TLS version and cipher suite do you get when you connect from `openssl`?
5. Force a different TLS version using `openssl`. Show your command and the output.
6. Force a different cipher suite using `openssl`. Show your command and the output.
7. (Optional) If you are able to leave your web server running, give me the URL so I can try it out myself